P08

De Bruijn Graphs for Short-Read Next-Generation Sequencing

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Domain DOM_CS

Abstract

Velvet applies graph simplification algorithms (tip clipping, bubble popping, and tour unrolling) to de Bruijn graphs generated from high-coverage short read datasets, producing contiguous contigs at draft genome scales.

Abstract quoted from D. R. Zerbino et al., Genome Research (2008).

Notes: record the page or section for every claim you make about this paper.

Reading prompts

1. What is the core problem addressed and what assumptions are made?
2. What mathematical or structural abstractions represent the physical phenomena?
3. How can this concept be mapped or transferred to analogous problems in other domains?